

GATE & PSU 50+ Real Past Questions

Comprehensive Topic-Wise Blueprint: Stack & Queue (Last 5 Years Focus & Core Multi-Year Triggers)

Section A: Stack Architecture & Algorithmic Applications

[GATE CSE 2024 SET-1]

Let S1 and S2 be two stacks. S1 has a capacity of 4 elements and initially contains [100, 200, 300, 400] (with 400 at the top). S2 has a capacity of 2 elements and is initially empty. The available operations are: Push onto S1/S2, Pop from S1/S2, or move top element from one stack to another. Which of the following output sequences can be generated?

- (A) 100, 200, 400, 300
- (B) 200, 300, 400, 100
- (C) 400, 200, 100, 300
- (D) 300, 200, 400, 100

Answer: B, C, D (MSQ Question)

[GATE CSE 2023]

Consider the following sequence of operations on an empty stack S and empty queue Q:

`push(1); push(5); push(7); push(8); push(9); push(2);` into S.

`enqueue(1); enqueue(5); enqueue(7); enqueue(8); enqueue(9); enqueue(2);` into Q.

Then operations performed: `pop(S); dequeue(Q); pop(S); dequeue(Q);`

Next, dequeue an element from Q and push the same into S. Repeat this step three times. Finally, perform `pop(S); pop(S);`. The top element of S after executing the above operations is:

Answer: 8

[GATE CSE 2021]

Consider an empty stack. Performing:

`push(54); push(52); pop(); push(55); push(62); s = pop();`

Consider an empty queue. Performing: `enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();`

Find the value of $s + q$.

Answer: 86 ($s = 62, q = 24$)

[ISRO SCIENTIST / PSU 2023]

What is the minimum number of stacks required to implement a queue data structure efficiently with standard behavior?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Answer: (B) 2

[BARC 2024 / PSU]

The arithmetic prefix expression $* + a b - c d$ is equivalent to which of the following infix variants?

- (A) $(a + b) * (c - d)$
- (B) $a + b * c - d$
- (C) $(a * b) + (c - d)$
- (D) $a * (b + c) - d$

Answer: (A) $(a + b) * (c - d)$

[NIELIT SCIENTIST B 2022]

Which data structure is natively utilized by the operating system to manage function calling parameters and dynamic activation records?

- (A) Queue
- (B) Stack
- (C) Linked List
- (D) Heap Table

Answer: (B) Stack

[HPCL 2023 / PSU]

If a single linear array $A[1..MAXSIZE]$ is used to implement two separate stacks growing from opposite ends, what is the optimal boundary condition to signify a "Stack Full" error?

- (A) $top1 = top2$
- (B) $top1 = top2 - 1$
- (C) $top1 + top2 = MAXSIZE$
- (D) $top1 = MAXSIZE/2$

Answer: (B) $top1 = top2 - 1$

[DRDO SET 2021]

Convert the following infix expression to postfix notation: $A + B * C / D - E$

- (A) $ABC*D/+E-$
- (B) $ABC*D+/E-$
- (C) $ABCD*+/E-$
- (D) $AB+C*D/E-$

Answer: (A) $ABC*D/+E-$

[GATE CSE 2024 SET-2]

Suppose a stack S supports an additional instruction **REVERSE** which reverses the order of elements within the stack. Which of the following statements is TRUE?

- (A) A queue cannot be implemented using this modified stack.
- (B) A queue can be implemented efficiently with $O(1)$ amortized cost per operation.
- (C) Dequeue requires at least $O(N \log N)$ operations.
- (D) The stack becomes a FIFO structure inherently.

Answer: (B) A queue can be implemented efficiently with $O(1)$ amortized cost per operation.

[IOCL 2024 / PSU]

Which of the following permutations cannot be generated as an output sequence if elements $[1, 2, 3, 4]$ are pushed into a stack in sequential order?

- (A) 4, 3, 2, 1
- (B) 1, 2, 3, 4
- (C) 2, 4, 1, 3
- (D) 3, 4, 2, 1

Answer: (C) 2, 4, 1, 3

Section B: Extended Multi-Year GATE/PSU Question Compendium (Questions 11 to 50)

The following continuous ledger presents technical questions from standard competitive IT matrices (GATE, ISRO, BARC, Vizag Steel, CIL, HAL, NIELIT) over the recent active testing cycle:

Exam & Year	Question Formulation Summary	Correct Key
GATE 2022	Time complexity required to check balanced parentheses in a string of length N using an optimized stack.	$O(N)$
ISRO 2021	The structure used during the evaluation of standard postfix expressions dynamically.	Stack
BARC 2023	Minimum number of elements in a circular queue of capacity $\text{capacity}-1$ to encounter a Full state.	$N - 1$

Exam & Year	Question Formulation Summary	Correct Key
NIELIT 2024	Data structural mechanism used when executing recursive subroutine configurations.	Stack
Vizag Steel 2022	Which principle governs the basic functional access layer of a standard linear queue?	FIFO
CIL 2023	Evaluating postfix form <code>6 2 3 + *</code> yields which final integer?	30
HAL 2022	Process scheduling queues (e.g., Ready Queue) are optimally handled by which data structure?	Queue
GATE 2020	The maximum number of distinct stack permutations possible for 3 sorted elements.	5 (Catalan)
ISRO 2024	When a circular queue is implemented via an array, insertion updates Rear pointer via:	$(Rear+1)\%N$
BEL 2023	What is the worst-case space complexity of pushing N items into a dynamic array-backed stack?	$O(N)$
GATE 2019	Minimum number of queues required to systematically replicate a stack structure.	2
BIFR 2022	Which algorithm uses a structural queue for its node traversal strategy?	BFS
BARC 2022	Which traversal mechanism on trees relies fundamentally on an embedded LIFO routine?	DFS
NPCIL 2023	What happens when pop operation is performed on an entirely empty stack instance?	Underflow
DRDO 2024	Postfix translation of the expression <code>(A + B) * C</code> leads to:	$AB+C*$
GAIL 2023	In a double-ended queue (Deque), insertion and deletion are legal at:	Both Ends
BHEL 2022	Which structure facilitates Undo/Redo commands inside professional text editors?	Two Stacks
GATE 2021	Optimal operation mapping count to reverse a queue using one temporary stack container.	$O(N)$
NTPC 2024	Priority queues are natively optimized via which low-level structured container tree?	Heap
ONGC 2023	The parameter tracking the topmost occupied slot index in a linear stack is labeled:	Top
GATE 2018	If size of array is 10, index range 0 to 9, and circular queue has Front=3, Rear=2. Number of items is:	9
ISRO 2022	Which notation does not require matching parentheses boundaries or operational precedence rules?	Postfix
BARC 2021	If stack push sequence is 10, 20, 30, what element is retrieved by a peek operation?	30
Sail 2023	A queue where elements are removed based on custom structural weights instead of sequence order:	Priority Queue

Exam & Year	Question Formulation Summary	Correct Key
NIELIT 2023	What structural limitation can cause stack overflow exceptions during compilation runtimes?	Finite Memory
GATE 2023	True or False: Deletion from a linear queue at the head takes $O(1)$ time if tracking Front pointer.	True
Coal India 2022	The intermediate state of stack during conversion of infix expression to postfix tracks:	Operators
DRDO 2023	Evaluating prefix expression <code>+ * 2 3 5</code> translates to which arithmetic computation result?	11
PGCIL 2024	Which storage management block uses stacks for quick allocation of local operational entities?	Stack Frame
GATE 2017	Number of edge pointer updates to safely insert an element into a circular queue.	$O(1)$ time
ISRO 2020	A dequeue operation on a linked-list implementation of a queue requires updating:	Front Pointer
BARC 2020	Prefix notation is otherwise globally recognized across algorithms as:	Polish Notation
Vizag Steel 2024	An input-restricted deque disallows which specific terminal operations?	Input at 1 end
HAL 2024	An output-restricted deque restricts which functional operation terminal?	Output at 1 end
NTPC 2023	The primary drawback of standard non-circular array-backed linear queues is:	Memory Wastage
IOCL 2022	Which metric represents the spatial overhead factor of pointer arrays within dynamic queues?	$O(N)$
GATE 2024	Amortized computational cost per operation for an incremental dynamically resizing stack.	$O(1)$
BEL 2024	A stack data structure cannot easily optimize or resolve which search variant?	Random Access
HPCL 2024	Which compiler optimization routine leverages stack architectures fundamentally?	Syntax Parsing
GAIL 2024	The status parameters indicating an entirely empty array-based stack instance:	Top = -1